AI.303

Swarm Robotics

COURSE OVERVIEW

Collective robotics: decentralized coordination, emergence, and scalable multi-robot behaviors.



ILLUSTRATIVE HERO IMAGERY

SWARM ROBOTICS

1

SESSION

60 MINUTES

ORIENTATION: STRENGTH IN NUMBERS

KEY TOPICS

- What is swarm robotics
- Decentralization and emergence
- Bio-inspiration: ants, bees
- Swarm advantages

TEACHING FOCUS

Many simple robots can do complex things.

▲ KEY TAKEAWAY Collective intelligence beats individual complexity.

SWARM ROBOTICS

2

SESSION

60 MINUTES

SWARM BEHAVIORS

KEY TOPICS

- Aggregation and dispersion
- Flocking and alignment
- Foraging
- Synchronization

TEACHING FOCUS

Program collective motion.

▲ **KEY TAKEAWAY** Simple local rules yield global patterns.

SWARM ROBOTICS

3

SESSION

60 MINUTES

LOCAL COMMUNICATION

KEY TOPICS

- Direct and indirect communication
- Stigmergy
- Local sensing
- Limited-range interaction

TEACHING FOCUS

Coordinate with only local knowledge.

▲ **KEY TAKEAWAY** Locality scales to large swarms.

SWARM ROBOTICS

4

SESSION

60 MINUTES

TASK ALLOCATION

KEY TOPICS

- Dynamic task partitioning
- Market-based and threshold methods
- Adaptivity
- Load balancing

TEACHING FOCUS

Divide work among robots.

KEY TAKEAWAY Tasks should find the right robot.

SWARM ROBOTICS

5

SESSION

 60 MINUTES

SWARM INTELLIGENCE ALGORITHMS

KEY TOPICS

- Ant colony optimization
- Particle swarm optimization
- Swarm applications
- Optimization

TEACHING FOCUS

Solve problems with swarms.

▲ **KEY TAKEAWAY** Swarm intelligence optimizes through collective search

SWARM ROBOTICS

6

SESSION

60 MINUTES

SWARM SYSTEMS & REVIEW

KEY TOPICS

- Swarm demo (simulation)
- Course recap and assessment
- Toward ROS (AI.304)
- Scalability discussion

TEACHING FOCUS

Coordinate many robots as one system.

KEY TAKEAWAY A swarm is more than the sum of its robots.